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This statement outlines my work related to the relative Langlands program. For a summary of my work on affine Springer fibers, Hessenberg varieties, and the combinatorics of torus links, see my webpage, linked here.

1 Background, introduction, and philosophy

My research focuses on affine Springer fibers, Hitchin systems, and their generalizations, especially those related to the relative Langlands program. More specifically, I study a *relative duality* between Dolbeault geometrizations of periods of automorphic forms and special values of L functions [HLM25] and applications of relative Hitchin systems to *relative endoscopy* for symmetric spaces [HL26].

For a reductive group G, the moduli space of G Higgs bundles Higgs_G digests the deep structure present in the Langlands program into their combinatorial or invariant theoretic shadows. Indeed, Hitchin constructed a remarkable fibration f_G generalizing the characteristic polynomial of a matrix which maps Higgs_G to an affine space and whose generic fibers are essentially abelian varieties [Hit87]. Through the Hitchin map, the geometry of Higgs_G is intimately tied to the *invariant theory* of the adjoint action of G on its Lie algebra $\mathfrak{g}$ in the sense of [MFK94], which was studied by [Kos63]. For example, taking the limit of the geometric Langlands conjectures[1] gives a duality of abelian varieties between generic fibers of f_G and generic fibers of $f_{\widehat{G}}$, where $\widehat{G}$ is the Langlands dual group of G [HT03, DP12, CZ17]. This in turn can be proven using an invariant theoretic manifestation of the duality of root systems between G and $\widehat{G}$.[2] In another direction, the Fundamental Lemma of Langlands and Shelstad becomes a computation of the cohomology of fibers of f_G, which is amenable to powerful algebro-geometric techniques in perverse sheaves. Ngô ultimately proved this by leveraging structure for the fibration f_G paralleling that of an invariant theoretic package for the Lie algebra $\mathfrak{g}$ [Ngo06, Ngo10].

I study Lagrangians Higgs_X in Higgs_G indexed by smooth, affine, spherical varieties $X = G/H$. Higgs_X arises naturally from the *relative Langlands program* as a semiclassical limit of the (global, de Rham) period sheaf [BZSV24], and fibers of the Hitchin map for Higgs_X give global models for *period integrals*. These Lagrangians are of interest in mirror symmetry as well, where they are examples of *BAA branes* and serve as examples of boundary conditions in dual field theories, see [Gai18].

If Higgs_G transforms problems in the Langlands program into invariant theory for G acting on $\mathfrak{g}$, then Higgs_X transforms problems in the relative Langlands program into invariant theory for the action of G on the cotangent bundle T^*X. For example, with Morrissey, I show that the geometry of Higgs_X for X a symmetric space is controlled over a large open locus by the geometry of the nilpotent cone of X, as defined and studied in [KR71, Sek84, Lev07]. In recent work with Leslie, we use this geometry to prove a *relative Fundamental Lemma*, an essential ingredient in stabilizing the relative trace formula, for the unitary symmetric space $X = \mathrm{U}_{2n}/\mathrm{GL}_{n,E}$ [HL26]. This is the first such case proved by geometry, and our method is general enough to recover the unitary Jacquet-Friedberg case proved via p-adic methods in [Les25].

It is fascinating that in the relative case, information flows in both directions between geometry and invariant theory. With Morrissey and Z. Luo [HLM25], I formulate the semiclassical limit of the global conjectures of [BZSV24]. Our work not only produces a natural dual object, situating a decades-old conjecture of Hitchin [Hit16] in the relative Langlands program, but also produces new relationships between the G orbits in T^*X of maximal dimension and the representation theory of the dual Hamiltonian. Under some hypotheses on X, this can even be used to conjecturally *define* the dual Hamiltonian associated to T^*X, whose definition in [BZSV24] is by its authors' account still *ad hoc*.

In the remainder of this statement, we comment on the projects above and future work planned in each direction. Namely, in §3, we discuss the relative Langlands program and its relationship to my work [HLM25]. In §2, we discuss the application to endoscopy for symmetric varieties, which includes the results of [HL26].

2 Endoscopy for symmetric varieties

The Fundamental Lemma of Langlands and Shelstad is an identity of *orbital integrals* which is essential to the stabilization of Arthur's trace formula. (See [Lan01] for an exposition.) It says roughly that

[1] which were recently proven in a series of papers [GR24a, ABC$^+$24a, CCF$^+$24, ABC$^+$24b, GR24b].

[2] This manifestation is the "universal centralizer," which by work of Donagi and Gaitsgory [DG02] and Knop [Kno96] is equivalent to the root system for G.

certain linear combinations of integrals over *rational* conjugacy classes can be expressed as integrals over a *geometric* conjugacy class for a loosely related *endoscopic group*. For example, when $F = \mathbb{R}$ is the field of real numbers and $G = \mathrm{SL}_2$, we can identify its Lie algebra with affine space $\mathfrak{g} \simeq \mathbf{A}^3$. The set of $\mathrm{SL}_2(\mathbb{C})$ conjugacy classes are the hyperbolas

$$-\det \begin{pmatrix} x & y \\ z & -x \end{pmatrix} = x^2 + yz = 0,$$

and each hyperbola splits into 2 real orbits $\mathcal{O}_+, \mathcal{O}_-$ depending on the sign of x. The fundamental lemma in this case says that for any compactly supported function the difference between integrating over $\mathcal{O}_+$ and $\mathcal{O}_-$ can be expressed as an integral of a related function over a torus.

A theory of endoscopy for symmetric spaces was developed by Spencer Leslie, who proved a fundamental lemma for the symmetric space $\mathrm{U}_{2n}/\mathrm{U}_n \times \mathrm{U}_n$ using p-adic harmonic analysis in [Les25] and produced a series of conjectures on what endoscopy should look like for general symmetric spaces in [Les24] and for some more specific cases in [LXZ25b, LXZ25a]. The resulting conjectures are essential to stabilizing the *relative trace formula* of Jacquet [JL85], which relates periods of automorphic forms to relative orbital integrals. The relative trace formula is a powerful tool for studying properties of *distinction* for representations of p-adic groups, and therefore has wide applications in the Langlands program.

To state the main result, we recall some notation. Let $X = G/H$ be a symmetric space, meaning that X is equipped with an involution θ and $(G^\theta)^\circ \subset H \subset G^\theta$. Let $\mathfrak{h}^\perp$ denote the (-1) eigenspace of θ acting on $\mathfrak{g}$. Then the adjoint action of H restricts to an action on $\mathfrak{h}^\perp$, and we put $\mathfrak{c} = \mathfrak{h}^\perp /\!\!/ H = \mathrm{Spec} k[\mathfrak{h}^\perp]^H$ for the GIT quotient parametrizing closed H orbits in $\mathfrak{h}^\perp$.

Consider $X = \mathrm{U}_{2n}/\mathrm{GL}_{n,E}$. For any n tuple κ with entries ± 1, we associate an endoscopic symmetric space

$$X_\kappa = X_r^{\mathrm{S}} \times X_s^{\mathrm{U}} = \left(\mathrm{U}_{2r}/\mathrm{GL}_{r,E}\right) \times \left(\mathrm{U}_{2s}/\mathrm{U}_s \times \mathrm{U}_s\right)$$

(with κ having r positive entries) with transfer map $\nu_\kappa \colon \mathfrak{c}_\kappa \to \mathfrak{c}$. We furthermore have *orbital integrals* $\mathrm{Orb}^\kappa_{X,a}$ at $a \in \mathfrak{c}(F)$, indexed by tuples κ, which integrate over linear combinations of orbital integrals for orbits over a with coefficients determined by κ. When $\kappa = 1$, all signs are positive and we call this the *stable orbital integral* $\mathrm{SOrb}_{X,a}$.

In my recent work joint with Leslie, we use *geometry* to deduce the following new fundamental lemma.

Theorem 1. *([HL26, Theorem 7.1]) Let E/F be an unramified quadratic extension of local fields, and consider the symmetric space $X = \mathrm{U}_{2n}/\mathrm{GL}_{n,E}$. Then, for any $a \in \mathfrak{c}(F)$ and $a_\kappa \in \mathfrak{c}_\kappa(F)$ with $\nu_\kappa(a_\kappa) = a$, we have*

$$\mathrm{Orb}^\kappa_{X,a} = \Delta(a)\mathrm{SOrb}_{X_\kappa, a_\kappa} \tag{2.1}$$

for an explicit transfer factor Δ.

This case arises in [LXZ25a], and interestingly, Leslie's original case $\mathrm{U}_{2n}/\mathrm{U}_n \times \mathrm{U}_n$ [Les25] appears as an endoscopic term in our formula. Our proof is the first that applies geometric techniques to prove an endoscopic fundamental lemma for symmetric spaces.

The proof of Theorem 1 follows the framework of geometric fundamental lemmas in [LN08, Ngo10, Yun16, Wan24, WZ26]. Namely, by standard reductions, one can reduce to proving Theorem 1 under the assumption that $F = \mathbb{F}_q(\!(\varpi)\!)$ is the field of Laurent series over a finite field k, and it suffices to prove (2.1) after evaluating both sides at the indicator function $\mathbf{1}_{\mathfrak{h}^\perp(k[\![\varpi]\!])}$. Under these simplifications, the transfer factor Δ becomes an explicit power of q.

The simplified integrals on each side of (2.1) are equal to point counts on *relative affine Springer fibers (RASF's)* for X and X_κ, respectively. We introduce these varieties as certain moduli spaces of *Hermitian self-dual* lattices which are fixed by an explicit involution. The cases of X and X_κ are distinguished by how the fixed involution and Hermitian form interact. It turns out that these RASF's are representable by locally finite type, projective schemes, and that, via an intricate argument modeled on the work of Jingren Chi, $\mathcal{M}^\star(a)$ is equidimensional.

The Grothendieck-Lefschetz trace formula allows us to reinterpret Theorem 1 as a computation of cohomology of RASF's. To compute the cohomology of RASF's, we globalize these moduli spaces using Higgs_X. We prove a product formula that relates point counts on Hitchin fibers in Higgs_X to point counts of products of associated RASF's. In particular, one can pass information back and forth between the local

and global moduli spaces, and Theorem 1 becomes a question of computing the cohomology of Higgs_X. We obtain analoguous results for X_κ.

For both X and X_κ, there are Hitchin-type maps $f_X\colon \mathrm{Higgs}_X \to \mathcal{A}_X$ and $f_\kappa\colon \mathrm{Higgs}_{X_\kappa} \to \mathcal{A}_\kappa$ where $\mathcal{A}_X$ and $\mathcal{A}_\kappa$ is an affine space. After an étale base change which we omit here,[3] there is a natural decomposition of the cohomology $\mathrm{R}f_{X,*}\overline{\mathbb{Q}}_\ell$ into κ constituent pieces, i.e. $\mathrm{R}f_{X,*}\overline{\mathbb{Q}}_\ell = \bigoplus_\kappa \left(\mathrm{R}f_{X,*}\overline{\mathbb{Q}}_\ell\right)_\kappa$. We call the case $\kappa = 1$ the stable constituent and denote it by $(\mathrm{R}f_{X,*}\overline{\mathbb{Q}}_\ell)_{\mathrm{st}}$. We prove the following global result by combining Ngô's support theorem and results of [HM25].

Theorem 2. *([HL26, Theorem 9.1]) There are canonical closed embeddings on Hitchin bases $\nu_\kappa\colon \mathcal{A}_\kappa \to \mathcal{A}$, and for any κ, we have an isomorphism of perverse sheaves over $\overline{k}$*

$$\nu_{\mathfrak{e}}^*\left(\mathrm{R}f_{X,*}\overline{\mathbb{Q}}_\ell\right)_\kappa[2r_\kappa(D)](r_\kappa(D)) = (\mathrm{R}f_{\kappa,*}\overline{\mathbb{Q}}_\ell)_{\mathrm{st}}$$

which induces an isomorphism of semisimplifications over k.

The proof of Theorem 2 reflects a delicate balance that has to be struck to make use of existing support theorems. Our future work proposes to test some of the bounds of these arguments to open up new cases; we meditate on some of these directions now.

One of the most fundamental difficulties in adapting the geometric machinery is the non-temperedness of X. We illustrate this with an example. Consider the case of $X = \mathrm{GL}_{2n} / \mathrm{Sp}_{2n}$. Work of Hitchin and Schaposnik prove that there is a family of curves $\mathscr{C} \to \mathcal{A}_X$ such that the fiber of the Hitchin map at $a \in \mathcal{A}_X$ is a compactification of the space of rank 2 vector bundles on $\mathscr{C}_a$ [HS14]. When $\mathscr{C}_a$ is smooth, as it is generically, the fiber is simply $\mathrm{Bun}_{\mathrm{GL}_2}(\mathscr{C}_a)$. The proof of Theorem 2 relies on Ngô's support theorem, a statement that is incredibly powerful, but is adapted specifically to the case of *abelian* fibrations. We propose to study this example, which has a particularly simple classification of supports.

Project 3. Show that in the case $X = \mathrm{GL}_{2n} / \mathrm{Sp}_{2n}$, when Higgs_X is the natural compactification of the family $\mathrm{Bun}_{\mathrm{GL}_2}(\mathscr{C}/\mathcal{A}_X)$, that all simple constituents of $\mathrm{R}f_*\overline{\mathbb{Q}}_{\ell,X}$ have full support.

There is a naive way to bound the supports using the action of the center of the centralizers, which is abelian. However, this does not constrain the supports enough to rule out smaller supports. Instead, we hope to apply more general support theorems, such as that of Migliorini and Shende [MS18].

We note that in the nontempered case, the fundamental lemma for the unit element may not imply the result for the full Hecke algebra. Hence, even after a support theorem like Project 3 is proved, it is likely that multiplicative methods similar to those carried out by [Wan24, WZ26] will be needed for the full range of arithmetic applications desired.

In another direction, we wish to study periods of more general spherical varieties. In this case, the most relevant obstruction is the failure of flatness of the Hitchin morphism. In particular, this failure reflects the fact that we do not yet know a dimension formula for such relative affine Springer fibers. In [Yun16], Yun considers such a case by using a unitary version of $X = \mathrm{GL}_n \times \mathrm{GL}_{n+1} / \mathrm{GL}_n$ by replacing Ngô's support theorem with semismallness of the Hitchin map. This semismallness is a reflection of the fact that the generic stabilizers of $H = \mathrm{GL}_n$ acting on $\mathfrak{h}^\perp$ are trivial, which is referred to as the condition that X be *strongly tempered*.

Project 4. Consider orbital integrals of more general spherical varieties. More explicitly:

1. For strongly tempered, smooth, affine spherical variety X with nontrivial dual symplectic representation S_X, adapt Yun's argument to deduce Fundamental Lemmas. For example, a unitary version of the Gan-Gross Prasad period for $X = \mathrm{SO}_{2n} \times \mathrm{SO}_{2n+1} / \mathrm{SO}_{2n}$ could be related to a non-unitary such period.

2. For general tempered X, unify semi-smallness with Ngô's support theorem.

Some evidence for the second proposed direction above comes from the factorization of the Hitchin map over the regular locus: In [HM25], we show that there is a large open set $\mathrm{Higgs}_X^{\mathrm{reg}} \subset \mathrm{Higgs}_X$ which factors $\mathrm{Higgs}_X^{\mathrm{reg}} \to \mathcal{A}_X^{\mathrm{reg}} \to \mathcal{A}_X$, where the latter map is an étale cover. If one could extend this factorization so

[3] We must restrict to a suitable elliptic locus and base change to a cameral cover.

the latter map is semismall and the former map captures the full cohomology of the fibers, Project 4 would be realized.

In another direction, we may hope to approach Project 4 using p-adic integration, which replaces the structure of perverse sheaves and the support theorem with more elementary integration along fibers. Such methods were used successfully to re-prove the Fundamental Lemma for Lie algebras and to extract deeper invariants from the Hitchin system to verify conjectures in mirror symmetry [GWZ20].

We also believe that our present arguments should be able to deduce cases that fall outside of the traditional boundaries of the relative Langlands program. We wish to consider one particular natural case of interest.

Project 5. Prove an endoscopic Fundamental Lemma for the case $X = \mathrm{SL}_n / \mathrm{SO}_n$. In particular, relate the corresponding endoscopic terms to the generalized dual Hamiltonian.

The case $X = \mathrm{SL}_n / \mathrm{SO}_n$ is a spherical variety with type N roots, meaning that its centralizers are disconnected. On the dual side, this reflects the fact that periods should be related to L functions attached to metaplectic groups–see for example [Jac91, CO12]–and that the dual Hamiltonian carries stacky structure. However, its relative Hitchin fibration is perfectly well behaved, and so is well suited to prove formulas involving orbital integrals. I am actively considering this case.

Finally, I will end this section with an easier example, which I expect to follow from similar methods to those used in [HL26]. This project would be a wonderful opportunity for a collaboration motivated graduate student interested in learning how to apply Hitchin systems to prove fundamental lemmas.

Project 6. Prove an endoscopic Fundamental Lemma for the case $X = \mathrm{U}_{2n+1}/\mathrm{U}_n \times \mathrm{U}_{n+1}$.

While this case is tempered and looks nominally similar to the Leslie's unitary Friedberg Jacquet case, it behaves geometrically more like a bridge case between tempered and nontempered symmetric spaces. In particular, naive arguments do not give a dimension formula for relative affine Springer fibers in this case as the notions of "smooth fibers" for $f\colon \mathrm{Higgs}_X \to \mathcal{A}_X$ and for $f_G\colon \mathrm{Higgs}_G \to \mathcal{A}_G$ are not quite compatible. However, this is still very manageable, as there is still generic conditions that bound the perverse behavior of relative affine Springer fibers and the relative Hitchin morphism is still flat.

3 Relative Langlands and the semiclassical limit

The relative Langlands program is fundamentally about matching specific measurements from each side of the Langlands correspondence. More specifically, to a Hamiltonian G variety M, one can associate a *period integral* and an L *function*. Ben-Zvi, Sakellaridis, and Venkatesh [BZSV24], building a wide literature of examples and precursors (see for example [SW22, Sak12, SV17]), propose a class $\mathcal{C}_G$ of Hamiltonian G varieties M, those that are *hyperspherical*, *anomoly-free*, *with no type N roots*, for which they can construct a dual object $\widehat{M} \in \mathcal{C}_{\widehat{G}}$. They conjecture that the period of M in $\mathcal{C}_G$ is equal to the central value of the L function of $\widehat{M}$ and vice-versa.

Higgs_X arises naturally when $M = T^*X$. By the function-sheaf dictionary of Grothendieck, to equate a period with an L function over the field of functions on C is the same as matching two sheaves on related geometric spaces. This takes the form of the *global conjectures* in [BZSV24], which relates a D module on the moduli of G bundles on C to a sheaf on the moduli of $\widehat{G}$ local systems. In [HLM25], we propose the following "semiclassical limit" of these conjectures when $M = T^*X$ is the cotangent bundle of a *tempered* spherical variety.

Conjecture 7. ([HLM25, Conjecture 7.2]) Assume that X is a tempered,[4] affine spherical variety with no type N roots. Then, over a generic locus in the Hitchin base, the Fourier-Mukai transform of the period sheaf $\mathcal{P}_M$ on Higgs_G is equal to the L sheaf $\mathcal{L}_{\widehat{M}}$ on $\mathrm{Higgs}_{\widehat{G}}$.

The period sheaf $\mathcal{P}_M$ is the associated graded of the period D module in [BZSV24], which is simply the structure sheaf of Higgs_X. The L sheaf $\mathcal{L}_{\widehat{M}}$, however, is obtained via a limiting process via nonabelian Hodge theory and carries with it much of the mystery of the dual Hamiltonian $\widehat{M}$. In particular, $\widehat{M}$

[4]Tempered here means that $\widehat{M}$ has trivial Arthur $\mathfrak{sl}_2$.

is built from the data of a dual group $\widehat{G}_X$ constructed in various guises in [Nad05, GN09, KS17, SV17] and a symplectic $\widehat{G}_X$ representation S_X. The representation S_X is subtle; in [BZSV24] they comment that its definition for them is *ad hoc*, and the only structural proof that it is symplectic is in [Les24] for symmetric spaces. The sheaf $\mathcal{L}_{\widehat{M}}$ is related to S_X via the *Dirac-Higgs bundle*, a construction from physics appearing in [Hit16]. For information on Dirac-Higgs bundles, see [FH24, Bla15] and for details on the "limit" calculations, see [HLM25, §7.4]. We note that Conjecture 7 can be seen as a precise formulation of predictions from mirror symmetry for Hitchin systems [KW07, GW09a, GW09b, Gai18, Hit16].

An incredible consequence of Conjecture 7 is that it unwinds to predict a new and simplified *definition* for the symplectic representation S_X in the case where X is tempered. (See [HLM25, Conjecture 7.8] and [Ham26].) This definition matches with that of [BZSV24] in all cases we have checked. Hence, one can observe from the invariant theory of the G action on T^*X exactly what the weights of S_X and hence the colors of $\widehat{M}$ should be.

In a general Hitchin fiber, Higgs_X is generically a union of translates of subabelian varieties of a Hitchin fiber. While we do not prove Conjecture 7 in full in [HLM25], we prove the following partial result.

Theorem 8. *([HLM25, Theorem 7.7]) Define S_X and $\widehat{M}$ using [Ham26] and [HLM25]. Then, Conjecture 7 holds "up to translations." That is, Conjecture 7 holds if we replace* Higgs_X *with a group of symmetries that acts simply transitively on it, and if we replace the Dirac-Higgs bundle with a pushforward of a trivial vector bundle.*

The distance between Conjecture 7 and Theorem 8 can be measured by M *Hecke operators*. Consider the correspondence stack Hecke_X which classifies tuples $((P,\Phi),(P',\Phi'),\xi)$ where $\xi\colon P \dashrightarrow P'$ is an elementary modification of P compatible with the Higgs fields Φ,Φ'. There are natural projections

$$\mathrm{Higgs}_X \xleftarrow{\pi_1} \mathrm{Hecke}_X \xrightarrow{\pi_2} \mathrm{Higgs}_X.$$

We have the M Hecke operator on coherent sheaves on Higgs_X given by $\mathcal{F} \mapsto \mathrm{R}\pi_{2,*}\mathrm{L}\pi_1^*(\mathcal{F})$. This has the effect of pulling back along translations between components of Higgs_X. On the dual side, we expect M Hecke operators to be compatible with tensoring by certain vector bundles on the dual. We call these latter operators $\widehat{M}$ Wilson operators. With Zeyu Wang, I plan to to address the following:

Project 9. Prove a categorical version of Conjecture 7 from Theorem 8. Specifically, show that the Fourier-Mukai transform intertwines M Hecke operators and $\widehat{M}$ Wilson operators to generate an equivalence of categories from Theorem 8.

We expect that Project 9 follows from a general philosophy that Conjecture 7 can be enhanced to a statement of modules over an L^2 duality. We propose this as a separate question within this project:

Question 10. Describe the endomorphism objects $\mathrm{End}_*(\mathcal{P}_M)$ and $\mathrm{End}_\otimes(\mathcal{L}_{\widehat{M}})$, taken in the convolution and tensor categories, respectively, in terms of the Hamiltonian varieties M and $\widehat{M}$, independent of polarization. Prove that these two sheaves are Fourier Mukai dual.

For the L sheaf, this is clear: We have an identification $\mathrm{End}\left(\bigwedge \mathrm{DH}(\widehat{G}_X, S_X^+)\right) = \mathrm{DH}(\widehat{G}_X, S_X)$. We hope that there is a similar description of $\mathrm{End}(\mathcal{P}_X)$ which is independent of the polarization X of $M = T^*X$.

We note here that the sheaf $\mathrm{DH}(\widehat{G}_X, S_X)$ carries a Clifford algebra structure which makes it an algebra object in the tensor category. Such a Clifford structure plays an important role in computing higher order derivatives of L functions [YZ17, YZ19, LW26], and is dual to the presence of nontrivial M Hecke operators. This is particularly intriguing as the nontrivial Hecke operators are the source of generic endoscopic terms for period integrals. We believe that these stories should be related, though precise statements are still in progress.

We note that a Fourier-Mukai equivalence of the algebras in Question 10 also gives a statement that does not require that the representation S_X be polarized. This happens in many examples of interest; for example the Gan-Gross-Prasad case $X = \mathrm{SO}_{2n} \times \mathrm{SO}_{2n+1} / \mathrm{SO}_{2n}$ for which the dual representation is the representation $\mathrm{std}_{2n} \otimes \mathrm{std}_{2n}$ as a representation of $\mathrm{SO}_{2n} \times \mathrm{Sp}_{2n}$.

In another direction, over positive characteristic the relative duality statement of Conjecture 7 has potential applications to the de Rham setting of [BZSV24]. We propose:

Project 11. In positive characteristic, generalize the arguments of [CZ17] to deduce a "generic" form of the global conjectures in [BZSV24].

The main idea of this direction is that the nonabelian Hodge correspondence, which connects between the Dolbeault and de Rham moduli spaces, can be made into an algebraic equivalence using the theory of Azumaya algebras. In particular, statements of duality between Higgs_G and $\mathrm{Higgs}_{\widehat{G}}$ can be transported to categorical statements of duality in de Rham geometric Langlands. This is particularly interesting as it is a *categorical* equivalence, not just a matching of sheaves.

Finally, we record a more speculative research direction. Consider the perverse cohomology sheaf ${}^p\mathscr{H}^k\big(\mathrm{R}f_{G,*}\overline{\mathbb{Q}}_\ell\big)$ as a D module on the Hitchin base, which carries a natural good filtration. A recent result of Maulik, Shen, and Yin proves that the associated graded of this filtration is the Fourier-Mukai dual of the sheaf $\Omega^k_{\mathrm{Higgs}_{\widehat{G}}}$ [MSY26].

Project 12. When $f\colon \mathrm{Higgs}_X \to \mathcal{A}_X$ is the relative Hitchin fibration associated to a spherical variety X, interpret the associated graded of ${}^p\mathscr{H}^k\big(\mathrm{R}f_*\overline{\mathbb{Q}}_\ell\big)$ in terms of the dual Hamiltonian.

We suspect that the answer is related to the Fourier-Mukai dual of $\bigwedge^k \mathrm{DH}(\widehat{G}_X, S_X^+)$. Resolving Project 12 may carry further implications for the theory of relative endoscopy, which we discuss in the next section. Namely, the supports of simple constituents in the perverse decomposition of $\mathrm{R}f_*\overline{\mathbb{Q}}_\ell$ controls the endoscopy for X. It would be conceptually satisfying to tie this decomposition to behavior of the dual Hamiltonian, and doing so may shed light on how to generalize the conjectures of [Les24] beyond the case of symmetric spaces.

References

[ABC+24a] Dima Arinkin, Dario Beraldo, Justin Campbell, Lin Chen, Joakim Faergeman, Dennis Gaitsgory, Kevin Lin, Sam Raskin, and Nick Rozenblyum, *Proof of the geometric Langlands conjecture II: Kac-Moody localization and the FLE*, 2024.

[ABC+24b] Dima Arinkin, Dario Beraldo, Lin Chen, Joakim Faergeman, Dennis Gaitsgory, Kevin Lin, Sam Raskin, and Nick Rozenblyum, *Proof of the geometric Langlands conjecture IV: ambidexterity*, 2024.

[Bla15] Jakob Lindblad Blaavand, *The Dirac-Higgs bundle*, Ph.D. thesis, University of Oxford, 2015, Available from http://people.maths.ox.ac.uk/hitchin/files/StudentsTheses/blaavand.pdf.

[BZSV24] David Ben-Zvi, Yiannis Sakellaridis, and Akshay Venkatesh, *Relative Langlands duality*, 2024, arXiv preprint, arXiv:2409.04677.

[CCF+24] Justin Campbell, Lin Chen, Joakim Faergeman, Dennis Gaitsgory, Kevin Lin, Sam Raskin, and Nick Rozenblyum, *Proof of the geometric Langlands conjecture III: compatibility with parabolic induction*, 2024.

[CO12] Gautam Chinta and Omer Offen, *Orthogonal period of a $GL_3(\mathbb{Z})$ Eisenstein series*, Representation theory, complex analysis, and integral geometry, Birkhäuser/Springer, New York, 2012, pp. 41–59. MR 2885075

[CZ17] Tsao-Hsien Chen and Xinwen Zhu, *Geometric Langlands in prime characteristic*, Compos. Math. **153** (2017), no. 2, 395–452. MR 3705230

[DG02] R. Y. Donagi and D. Gaitsgory, *The gerbe of Higgs bundles*, Transform. Groups **7** (2002), no. 2, 109–153. MR 1903115

[DP12] Ron Donagi and Tony Pantev, *Langlands duality for Hitchin systems*, Inventiones mathematicae **189** (2012), no. 3, 653–735.

[FH24] Emilio Franco and Robert Hanson, *The Dirac-Higgs complex and categorification of (BBB)-branes*, Int. Math. Res. Not. IMRN (2024), no. 19, 12919–12953. MR 4805839

[Gai18] Davide Gaiotto, *S-duality of boundary conditions and the Geometric Langlands program*, Proc. Symp. Pure Math, vol. 98, 2018, p. 139.

[GN09] Dennis Gaitsgory and David Nadler, *Hecke operators on quasimaps into horospherical varieties*, Doc. Math. **14** (2009), 19–46. MR 2505056

[GR24a] Dennis Gaitsgory and Sam Raskin, *Proof of the geometric Langlands conjecture I: construction of the functor*, 2024.

[GR24b] ———, *Proof of the geometric Langlands conjecture V: the multiplicity one theorem*, 2024.

[GW09a] Davide Gaiotto and Edward Witten, *S-duality of boundary conditions in $\mathcal{N} = 4$ super Yang-Mills theory*, Advances in Theoretical and Mathematical Physics **13** (2009), no. 3, 721–896.

[GW09b] ———, *Supersymmetric boundary conditions in* $\mathcal{N} = 4$ *super Yang-Mills theory*, Journal of Statistical Physics **135** (2009), no. 5, 789–855.

[GWZ20] Michael Groechenig, Dimitri Wyss, and Paul Ziegler, *Geometric stabilisation via p-adic integration*, Journal of the American Mathematical Society **33** (2020), no. 3, 807–873.

[Ham26] Thomas Hameister, *Note on symplectic representations and divisors*, 2026, Accessed at https://hameist.github.io/repsanddivisors.pdf.

[Hit87] Nigel Hitchin, *Stable bundles and integrable systems*, Duke Math. J. **54** (1987), no. 1, 91–114. MR 885778

[Hit16] ———, *Higgs bundles and characteristic classes*, Arbeitstagung Bonn 2013, Progr. Math., vol. 319, Birkhäuser/Springer, Cham, 2016, pp. 247–264. MR 3618052

[HL26] Thomas Hameister and Spencer Leslie, *Relative Hitchin fibrations and endoscopy for unitary Friedberg–Jacquet varieties*, arXiv preprint (2026).

[HLM25] Thomas Hameister, Zhilin Luo, and Benedict Morrissey, *Relative Dolbeault geometric Langlands via the regular quotient*, 2025.

[HM25] Thomas Hameister and Benedict Morrissey, *The Hitchin fibration for symmetric pairs*, Adv. Math. **482** (2025), Paper No. 110560, 68. MR 4966334

[HS14] Nigel Hitchin and Laura P. Schaposnik, *Nonabelianization of Higgs bundles*, J. Differential Geom. **97** (2014), no. 1, 79–89. MR 3229050

[HT03] Tamás Hausel and Michael Thaddeus, *Mirror symmetry, Langlands duality, and the Hitchin system*, Invent. Math. **153** (2003), no. 1, 197–229. MR 1990670

[Jac91] Hervé Jacquet, *Représentations distinguées pour le groupe orthogonal*, C. R. Acad. Sci. Paris Sér. I Math. **312** (1991), no. 13, 957–961. MR 1113084

[JL85] H. Jacquet and K. F. Lai, *A relative trace formula*, Compositio Math. **54** (1985), no. 2, 243–310. MR 783512

[Kno96] Friedrich Knop, *Automorphisms, root systems, and compactifications of homogeneous varieties*, J. Amer. Math. Soc. **9** (1996), no. 1, 153–174. MR 1311823

[Kos63] Bertram Kostant, *Lie group representations on polynomial rings*, American Journal of Mathematics **85** (1963), no. 3, 327–404.

[KR71] B. Kostant and S. Rallis, *Orbits and representations associated with symmetric spaces*, Amer. J. Math. **93** (1971), 753–809. MR 311837

[KS17] F. Knop and B. Schalke, *The dual group of a spherical variety*, Trans. Moscow Math. Soc. **78** (2017), 187–216. MR 3738085

[KW07] Anton Kapustin and Edward Witten, *Electric-magnetic duality and the geometric Langlands program*, Commun. Number Theory Phys. **1** (2007), no. 1, 1–236. MR 2306566

[Lan01] Robert P. Langlands, *The trace formula and its applications: an introduction to the work of James Arthur*, Canad. Math. Bull. **44** (2001), no. 2, 160–209. MR 1827854

[Les24] Spencer Leslie, *Symmetric varieties for endoscopic groups*, 2024.

[Les25] Spencer Leslie, *The endoscopic fundamental lemma for unitary Friedberg-Jacquet periods*, Ann. of Math. (2) **201** (2025), no. 3, 551–645. MR 4899538

[Lev07] Paul Levy, *Involutions of reductive Lie algebras in positive characteristic*, Adv. Math. **210** (2007), no. 2, 505–559. MR 2303231

[LN08] Gérard Laumon and Bao Châu Ngô, *Le lemme fondamental pour les groupes unitaires*, Ann. of Math. (2) **168** (2008), no. 2, 477–573. MR 2434884

[LW26] Shurui Liu and Zeyu Wang, *Higher period integrals and derivatives of L-functions*, 2026.

[LXZ25a] Spencer Leslie, Jingwei Xiao, and Wei Zhang, *Unitary Friedberg-Jacquet periods and their twists: Fundamental lemmas*, 2025.

[LXZ25b] ———, *Unitary Friedberg-Jacquet periods and their twists: Relative trace formulas*, 2025.

[MFK94] D. Mumford, J. Fogarty, and F. Kirwan, *Geometric invariant theory*, third ed., Ergebnisse der Mathematik und ihrer Grenzgebiete (2) [Results in Mathematics and Related Areas (2)], vol. 34, Springer-Verlag, Berlin, 1994. MR 1304906

[MS18] Luca Migliorini and Vivek Shende, *Higher discriminants and the topology of algebraic maps*, Algebr. Geom. **5** (2018), no. 1, 114–130. MR 3734111

[MSY26] Davesh Maulik, Junliang Shen, and Qizheng Yin, *Fourier-Mukai transforms and the decomposition theorem for integrable systems*, Algebr. Geom. **13** (2026), no. 2, 266–304. MR 5034710

[Nad05] David Nadler, *Perverse sheaves on real loop Grassmannians*, Invent. Math. **159** (2005), no. 1, 1–73. MR 2142332

[Ngo06] Bao Chau Ngo, *Fibration de Hitchin et endoscopie*, Invent. Math. **164** (2006), no. 2, 399–453. MR

2218781

[Ngo10] ———, *Le lemme fondamental pour les algèbres de Lie*, Publ. Math. Inst. Hautes Études Sci. (2010), no. 111, 1–169. MR 2653248

[Sak12] Yiannis Sakellaridis, *Spherical varieties and integral representations of L-functions*, Algebra Number Theory **6** (2012), no. 4, 611–667. MR 2966713

[Sek84] Jiro Sekiguchi, *The nilpotent subvariety of the vector space associated to a symmetric pair*, Publ. Res. Inst. Math. Sci. **20** (1984), no. 1, 155–212. MR 736100

[SV17] Yiannis Sakellaridis and Akshay Venkatesh, *Periods and harmonic analysis on spherical varieties*, Astérisque (2017), no. 396, viii+360. MR 3764130

[SW22] Yiannis Sakellaridis and Jonathan Wang, *Intersection complexes and unramified L-factors*, J. Amer. Math. Soc. **35** (2022), no. 3, 799–910. MR 4433079

[Wan24] X. Griffin Wang, *Multiplicative hitchin fibration and fundamental lemma*, 2024.

[WZ26] X. Griffin Wang and Zhiyu Zhang, *Relative fundamental lemmas for spherical Hecke algebras and multiplicative Hitchin fibrations: the Jacquet–Rallis case*, 2026.

[Yun16] Zhiwei Yun, *Lectures on Springer theories and orbital integrals*, arXiv preprint arXiv:1602.01451 (2016).

[YZ17] Zhiwei Yun and Wei Zhang, *Shtukas and the Taylor expansion of L-functions*, Ann. of Math. (2) **186** (2017), no. 3, 767–911. MR 3702678

[YZ19] ———, *Shtukas and the Taylor expansion of L-functions (II)*, Ann. of Math. (2) **189** (2019), no. 2, 393–526. MR 3919362